

Candidate Brief — Trading Results Analysis Challenge

Business Context

You've joined the Data Stream team as an Analytics Engineer at a global multi-asset brokerage. Your first task: prepare an analysis of trading results for management.

The leadership team wants to know:

- Who is performing best
- Who is underperforming
- How many traders are active
- Which clients/accounts are at risk
- Any other important insights you can uncover.

The dataset comes from production exports its raw and contains inconsistencies.

Your job is to model it into clean, analysis-ready marts and produce the requested analysis.

Data Provided (CSV format)

1. trades_raw trade records

Fields: trade_id, platform, account_id, client_external_id, symbol, side, volume, open_time, close_time, open_price, close_price, commission, realized_pnl, book_flag, counterparty, quote_currency, status

2. accounts_raw trading account registry

Fields: account_id, platform, client_id, base_currency, created_at, closed_at, salesforce_account_id, is_system, is_deleted

3. clients_raw client master

Fields: client_id, client_external_id, jurisdiction, segment, created_at

4. balances_eod_raw end-of-day balances

Fields: account_id, platform, date, balance, equity, floating_pnl, credit, margin_level

5. symbols_ref symbol normalization reference

Fields: platform, platform_symbol, std_symbol, asset_class, quote_currency, tick_value

Business Questions

Performance

- Top 10 performers and underperformers by net PnL (realized_pnl + commission), at client and symbol level.
- Which symbols contribute most to overall losses?

Activity

- Number of active traders (clients/accounts) per day and per week.
- Average trade size by symbol and by client segment.

Risk / Account Health

- Clients/accounts with the largest equity drawdown.
- Accounts marked is_deleted = true but still trading.

Modeling Expectations

- Normalize symbols using symbols_ref.
- Link clients across trades_raw and accounts_raw (some trades have missing client_external_id).
- Standardize segment values and validate jurisdiction codes.

Important: Prepare at least 3 data quality tests using dbt test (e.g., uniqueness, not nulls, accepted values, or custom business logic tests).

Deliverables

1. Data Marts (dbt):

- Dimensions: dim_client, dim_account, dim_symbol
- Facts: fact_trades, fact_account_eod, f_client_performance_daily

2. Analysis Output:

- Dashboard (Tableau, Power BI, Looker Studio, etc.) OR
- Jupyter Notebook / Python script / Excel / Google Sheets

Technical Requirements

- Use dbt to transform raw data into marts.

- Document assumptions and sign conventions.
- Keep code and files clean and well-structured.

Submission

- Git repo or zipped folder with dbt project, analysis output, summary insights, and run instructions.
- Include a README: how to run, approach, and challenges faced.